



HARDWARE SYLLABUS

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1. Introduction

- a. Objectives
- b. What is Computer?
 - i. Basic Applications of Computer
 - ii. types of computer
- c. Architecture of Computer System
 - i. Central Processing Unit

2. CPU Features

- Single-core
- Multicore
- Virtualization
- Hyper threading
- Speeds
- Overclocking
- Integrated GPU

3. Compatibility

- AMD
- Intel

4. Expansion Cards

- Video cards
- Onboard
- Add-on card
- Sound cards
- Network interface card
- USB expansion card

5. Mother Board

- ATX
- mATX
- ITX
- mITX

6. Motherboard connectors types

- PCI
- PCIe
- Riser card
- Socket types
- SATA
- IDE
- Front panel connector
- Internal USB connector

7. Keyboard, mouse and VDU

8. Other Input & output devices

9. Computer Memory

- **Memory System:**
 - Semiconductor RAM Memories, Internal Organization of Memory Chips, Static Memories, Asynchronous DRAMS, Synchronous DRAMSRAMBUS Memory.

- **Read-Only Memories:**
 - ROM, PROM, EPROM, EEPROM, Flash Memory, Speed, Size, and Cost of Memory.
- **Secondary Storage:**
 - Magnetic Hard Disks, Optical Disks, Magnetic Tape Systems

10. Concept of Hardware and Software

- Hardware
- Software
 - Application Software
 - Systems software

11. Concept of computing, data and information.

12. Practicals:

- Installation of any Operating system [Ex Windows 10]
- Types of software. System software OS, Compiler. Application software like MS office.
- Functions of an operating system. Disk operating system

- a. Concept of GUI, Modes of starting on different occasions.
- b. Desktop, Icon, selecting, choosing, drag and drop.
- c. My computer, network neighborhood / network places.
- d. Recycle bin, briefcase, task bar, start menu, tool bar, and menus.
- e. Windows Explorer.
- f. Properties of files and folders.
- g. Executing application programs.
- h. Properties of connected devices.
 - i. Applications under windows accessories.
 - j. Windows Help.
- k. Finding files, folders, computers.
- l. Control panel.
- m. Installed devices and properties.
- n. Utilities for recovering data from defective/bad hard disks.

13 .Partition

- Dynamic
- Basic
- Primary
- Extended
- Logical
- GPT
- How to Backup/Restore your Windows partition with the bootable image disk
- Duplicating a partition (creating a multi boot system)
- file permission and sharing

14. File system types/formatting

- ExFAT
- FAT32
- NTFS
- CDFS
- NFS
- ext3, ext4
- HFS
- Swap partition

15 . Basic Commands

- ipconfig
- ping
- tracert
- netstat
- nslookup
- shutdown
- dism
- sfc
- chkdsk
- diskpart